

A Spatial-Behavioural Logic for the Rho Combinators

Structural formulae, one context modality, and decidable occupancy of the expressiveness lattice

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Abstract

The companion note [7] repairs the name-free concurrent combinators for the rho calculus and replaces the separation theorem with an expressiveness lattice over subsets of the atom set. Each edge of that lattice is witnessed by an invariant — name closure, atom count, message count — established in the metatheory and stated in prose. This note supplies a logic in which those invariants are formulae. The logic is the usual propositional connectives, the spatial pair $\mathbf{0}$ and $|$, one structural production per combinator shape with name positions written $@\phi$, a single context modality $\langle\phi\rangle\psi$, and a parameterised least fixpoint over namespaces. It has no restriction connectives, no guarantee, and no name sort beyond $@\phi$, because the calculus has no binders and every name is a quotation.

Three things follow. Characteristic formulae exist and are exact, so the logic separates terms precisely up to structural congruence, and nominals, counting and the temporal operators are all derived rather than primitive (§4). Occupancy of a lattice point — the assertion that a term is a term of the sub-calculus over an atom set $\mathcal{A}$ — is the nullary fixpoint $\text{Occ}(\mathcal{A})$, is checkable in one pass over the term (Theorem 6.5), and is invariant under reduction exactly when $\mathcal{A}$ is rule-closed (Proposition 6.7). The constructor axis has a second and sharper formula, the namespace closure $\text{Cl}_{\mathcal{A}}(T)$, which is the fixpoint whose disjuncts are in bijection with the constructors present; it is the logical form of the name-closure lemma. Third, rule-closure has a consequence for the lattice as drawn: at the eight points carrying $\text{cons}_{\mathbf{d}}$ without $\mathbf{d}$, occupancy is not invariant, so the two axes of [7, Fig. 2] are not independent for this predicate (Proposition 6.8). The modality-free fragment is decidable outright; the full logic is decidable on the $\mathbf{e}$ -free sublattice modulo one finite-witness obligation, and undecidable above it.

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1 What this note is

The expressiveness lattice of [7, §10] is a family of sub-calculi of the rho combinators, ordered by inclusion of atom sets, with each strict edge witnessed by a behaviour available above it and not below. The witnesses are produced by three invariants: names reachable from a term lie in the closure of its names under the node shapes of the constructors present; every rule but e 's strictly decreases the count of non-message atoms; and d is the only free source of copies.

Those invariants are stated about the calculus. This note states them *in* a logic over the calculus — so that “ T sits at or below lattice point S ” is a formula, and so that checking it is a decision procedure rather than a proof. That is the whole of the brief. In particular:

- This is a logic, not a type system. Nothing here is a typing judgement, nothing is syntax-directed on a derivation, and there is no subject-reduction theorem for terms-in-context. The intended route to a type system is generation from the logic, which is out of scope here.
- The logic is deliberately intensional. Theorem 4.4 says it separates terms exactly up to $\equiv$, which is far finer than the equivalence $\approx_{\llbracket \mathcal{C}_{\text{rho}} \rrbracket}$ of [7, Thm. 7.5]. That is wanted for occupancy and is wrong for anything else; §7 says so plainly.
- Occupancy formulae must be decidable. They are, and better than decidable: linear (Theorem 6.5).

The logic is a near relative of the hybrid logic of the F1R3Space semantics note [5], and diverges from it in four places, each traceable to a feature of the target calculus rather than to taste: there is no restriction, so revelation, hiding, freshness and $\text{free}(a)$ all go; there is no unforgeable name, so the name sort collapses into $@(-)$ and needs no Boolean structure of its own (Proposition 3.5); the guarantee $\phi \triangleright_{\eta} \psi$ is absent, which is what keeps the structural fragment decidable; and the three action modalities are replaced by one context modality rather than by the four of [8]. The last is a restriction and is discussed in §7.

2 The calculus recalled

We follow [7, §3] exactly, and repeat only what the logic needs.

Definition 2.1 (Syntax).

$$\begin{aligned}
A &::= m(a, b) \mid d(a, b, c) \mid k(a) \mid fw(a, b) \mid b_l(a, b) \mid b_r(a, b) \mid s(a, b, c) \\
&\quad \mid e(a) \mid q(a, P) \\
&\quad \mid cons_l(a, b, c) \mid cons_m(a, b, c) \mid cons_d(a, b, c, e) \mid cons_s(a, b, c, e) \\
P, Q &::= \mathbf{0} \mid A \mid P \mid Q \qquad a, b, c ::= @P
\end{aligned}$$

Write $\mathcal{A}_{\text{full}}$ for the set of thirteen atom shapes, and $\text{ar}(\alpha)$ for the arity of α as a term former. Every argument of every shape is a name except the second argument of q , which is a process.

Definition 2.2 (Structural congruence). $\equiv$ is the least congruence making $(\text{Proc}, |, \mathbf{0})$ an abelian monoid. There are no binders, hence no α -equivalence, and no clause identifying a dropped quotation with its content.

Definition 2.3 (Components). $\text{comps}(P)$ is the multiset of atoms at the top level of P after erasure of $\mathbf{0}$; it is a complete invariant of $\equiv$, so $P \equiv Q$ iff $\text{comps}(P) = \text{comps}(Q)$ as multisets, with equality of atoms taken componentwise up to $\equiv_{\mathcal{N}}$ on names and $\equiv$ on the payload of q .

Proposition 2.4 (Name equivalence on the target is structural). *On combinator names, $@P \equiv_{\mathcal{N}} @Q$ iff $P \equiv Q$.*

Proof. The source clause $@*x \equiv_{\mathcal{N}} x$ has no instance here: $*(-)$ is not a process former of Definition 2.1, the drop having been demoted to the atom e [7, Rem. 3.2]. What remains of the definition of $\equiv_{\mathcal{N}}$ is the congruence generated by $\equiv$ under quotation. $\square$

Proposition 2.4 discharges corrigendum 4 of [7, §12], which observes that $\equiv_{\mathcal{N}}$ is used in the side condition on observation and never defined for the combinators. It also says that quotation is injective and onto the name universe, which is what §3 trades on and what the source calculus does not enjoy.

Definition 2.5 (Reduction).

$$\begin{aligned}
d(a, b, c) \mid m(a, v) &\longrightarrow m(b, v) \mid m(c, v) & b_r(a, b) \mid m(a, v) &\longrightarrow fw(b, v) \\
k(a) \mid m(a, v) &\longrightarrow \mathbf{0} & b_l(a, b) \mid m(a, v) &\longrightarrow fw(v, b) \\
fw(a, b) \mid m(a, v) &\longrightarrow m(b, v) & s(a, b, c) \mid m(a, v) &\longrightarrow fw(b, c) \\
e(a) \mid m(a, @P) &\longrightarrow P & fw(a, b) \mid q(a, P) &\longrightarrow m(b, @P) \\
cons_l(a, b, c) \mid m(a, @P) \mid m(b, @Q) &\longrightarrow m(c, @(P \mid Q)) \\
cons_m(a, b, c) \mid m(a, u) \mid m(b, v) &\longrightarrow m(c, @m(u, v)) \\
cons_d(a, b, c, e) \mid m(a, u) \mid m(b, v) \mid m(c, w) &\longrightarrow m(e, @d(u, v, w)) \\
cons_s(a, b, c, e) \mid m(a, u) \mid m(b, v) \mid m(c, w) &\longrightarrow m(e, @s(u, v, w))
\end{aligned}$$

closed under $|$ and under $\equiv$, with subjects matched up to $\equiv_{\mathcal{N}}$.

Every rule has between two and four premises, each premise an atom, and the premises of a rule share names. That is the only structural fact the modality of §3 uses.

3 The logic

3.1 Grammar

Definition 3.1 (Formulae). Two sorts. Name formulae (namespaces) ν , and process formulae ϕ :

$\nu ::=$	$@\phi \mid U$	namespace, namespace parameter
$\phi, \psi ::=$	$\text{true} \mid \neg\phi \mid \phi \wedge \psi \mid \phi \vee \psi \mid \phi \Rightarrow \psi$	propositional
	$\mid \mathbf{0} \mid \phi \mid \psi$	spatial
	$\mid \mathbf{m}(\nu_1, \nu_2) \mid \mathbf{d}(\nu_1, \nu_2, \nu_3) \mid \mathbf{k}(\nu_1)$	structural
	$\mid \mathbf{fw}(\nu_1, \nu_2) \mid \mathbf{b}_l(\nu_1, \nu_2) \mid \mathbf{b}_r(\nu_1, \nu_2)$	
	$\mid \mathbf{s}(\nu_1, \nu_2, \nu_3) \mid \mathbf{e}(\nu_1) \mid \mathbf{q}(\nu_1, \phi)$	
	$\mid \mathbf{cons}_l(\nu_1, \nu_2, \nu_3) \mid \mathbf{cons}_m(\nu_1, \nu_2, \nu_3)$	
	$\mid \mathbf{cons}_d(\nu_1, \nu_2, \nu_3, \nu_4) \mid \mathbf{cons}_s(\nu_1, \nu_2, \nu_3, \nu_4)$	
	$\mid \langle\phi\rangle\psi$	context modality
	$\mid \mu X(\vec{U}).\phi \mid X(\vec{v})$	fixpoint, application

In $\mu X(\vec{U}).\phi$ the parameters $\vec{U}$ are bound in ϕ , and X must occur positively in ϕ ; the application $X(\vec{v})$ must supply as many namespaces as X was declared with.

The structural productions are in bijection with the atom shapes of Definition 2.1, and each has the arity of its shape. The one asymmetry is $\mathbf{q}(\nu_1, \phi)$, whose second argument is a process formula. It is not a defect of the grammar but a faithful record of the calculus: $\mathbf{q}$ is the introduction form for a quotation, and its payload has not yet been quoted.

Remark 3.2 (The gate migrates a formula across the sorts). Under the release rule $\mathbf{fw}(a, b) \mid \mathbf{q}(a, P) \longrightarrow \mathbf{m}(b, @P)$ the payload is described by ψ before the step and by $@\psi$ after. Formally, $\mathbf{q}(\nu, \psi)$ is carried to $\mathbf{m}(\nu', @\psi)$ where ν' is the forwarder's target. The logic is thereby able to watch the sort change that quotation performs, and it is the only place in the grammar where that change is visible. A grammar that wrote $\mathbf{q}(\nu_1, \nu_2)$ would be uniform and would be uninhabited: no well-formed $\mathbf{q}$ term has a name in its second argument.

Decision 3.3 (No guarantee, no restriction, one modality). $\phi \triangleright_\eta \psi$ is omitted, and with it the quantification over an infinite class of environments that makes spatial model checking hard; the single-step consequence of it that this setting wants is recovered by $\langle\phi\rangle\psi$. The restriction connectives of [5] are omitted because the calculus has no *new*. The modality set of [8] is cut from four to one; §7 records the cost.

3.2 Satisfaction

Satisfaction is defined on closed terms and closed formulae, relative to an environment $\mathcal{E}$ interpreting free fixpoint variables. We suppress $\mathcal{E}$ except in the fixpoint clauses.

Definition 3.4 (Name sort).

$$a \models @\phi \quad \text{iff} \quad a \equiv_{\mathcal{N}} @P \text{ for some } P \text{ with } P \models \phi.$$

Proposition 3.5 (The name sort is definable). *Quotation is a bijection from $\equiv$ -classes of processes to $\equiv_{\mathcal{N}}$ -classes of names, and $\llbracket @(-) \rrbracket$ preserves and reflects Boolean structure:*

$$@(\phi \wedge \psi) \equiv @\phi \wedge @\psi, \quad @(\phi \vee \psi) \equiv @\phi \vee @\psi, \quad @(\neg\phi) \equiv \neg @\phi,$$

and $@\text{true}$ is the namespace of all names.

Proof. Bijectivity is Proposition 2.4 together with the fact that every name of Definition 2.1 is a quotation. The four equivalences follow by unfolding Definition 3.4 against that bijection. $\square$

So the name sort carries no productions of its own beyond $@(-)$ and the parameter U , and the Boolean namespace algebra that [5] must add to the name sort is here a theorem. The reason is exactly that the combinators have no unforgeable names: in [5] $a \models @\phi$ is false at every atom, which forces $\top_{\mathcal{N}}$ to be primitive and a separate Boolean algebra with it.

Definition 3.6 (Process sort). Propositional clauses are standard. Otherwise:

$$\begin{array}{ll}
P \models \mathbf{0} & \text{iff } P \equiv \mathbf{0} \\
P \models \phi \mid \psi & \text{iff } \exists P_1, P_2. P \equiv P_1 \mid P_2, P_1 \models \phi, P_2 \models \psi \\
P \models \alpha(\nu_1, \dots, \nu_n) & \text{iff } \exists a_1, \dots, a_n. P \equiv \alpha(a_1, \dots, a_n) \text{ and } a_i \models \nu_i \text{ for all } i \\
& \text{for each shape } \alpha \neq \mathbf{q} \text{ of arity } n \\
P \models \mathbf{q}(\nu, \psi) & \text{iff } \exists a, S. P \equiv \mathbf{q}(a, S), a \models \nu, S \models \psi \\
P \models \langle \phi \rangle \psi & \text{iff } \exists R \models \phi. \exists P'. P \mid R \longrightarrow_R P' \text{ and } P' \models \psi \\
P \models_{\mathcal{E}} \mu X(\vec{U}).\phi & \text{iff } P \in \bigcap \{ F : \Phi_{\mathcal{E}}(F) \subseteq F \} \\
P \models_{\mathcal{E}} X(\vec{v}) & \text{iff } P \in \mathcal{E}(X)(\llbracket \vec{v} \rrbracket)
\end{array}$$

where $\Phi_{\mathcal{E}}$ is the monotone operator on $(\mathcal{P}(\text{Name})^n \rightarrow \mathcal{P}(\text{Proc}))$ induced by ϕ , and the least fixpoint exists by Knaster–Tarski since X occurs positively.

Definition 3.7 (Marked reaction). $P \mid R \longrightarrow_R P'$ holds when there is a derivation of $P \mid R \longrightarrow P'$ by a single rule of Definition 2.5 in which *every* atom of $\text{comps}(R)$ is a premise of that rule.

Remark 3.8 (Why the marking). Without it, $P \mid R \longrightarrow P'$ is discharged by any internal step of P whatsoever, and the formula ϕ describing the context does no work: every term with a reduction would satisfy every $\langle \phi \rangle \psi$. With it, $\langle \mathbf{0} \rangle \psi$ is the internal-step modality — $\text{comps}(\mathbf{0})$ is empty, so the condition is vacuous and the whole redex comes from P — and $\langle \mathbf{m}(\nu_1, \nu_2) \rangle \psi$ says that P offers a consumer at a name in ν_1 willing to accept a payload in ν_2 . The condition is on the derivation and not on P' , which is where it has to be, since a residual carries no record of which premises produced it. This follows [8, Def. 5.4].

Remark 3.9 (One step, and only one). $\longrightarrow_R$ is a single rule application, not its reflexive transitive closure. Over many steps a context could supply an atom, take a name from P , and act on it, and the modality would be observing a protocol rather than a reaction. The multi-step reading is available as a derived operator (§4) and is the right thing to write when it is wanted.

Remark 3.10 (Bounded contexts, for free). Definition 3.7 bounds $\text{comps}(R)$ by the largest premise count of any rule, which is four; and if $\text{comps}(R)$ is nonempty, at least one premise must come from P unless R alone forms a redex. This is what makes the modal fragment tractable at all, and it is a gift of the combinator presentation: in a calculus with prefixing, a reacting context can be arbitrarily large.

3.3 Positivity

Proposition 3.11 (Polarity). $\phi \mid \psi$, $\alpha(\dots)$, $@(-)$ and $\langle \phi \rangle \psi$ are monotone in every argument. The negative positions are exactly the scope of $\neg$ and the antecedent of $\Rightarrow$.

Proof. Each clause of Definitions 3.4 and 3.6 other than $\neg$ and $\Rightarrow$ is a positive existential over the interpretations of its arguments. $\square$

Note that $\langle \phi \rangle \psi$ is monotone in ϕ as well as in ψ : the context is existentially quantified, so enlarging the class of admissible contexts can only enlarge the satisfying set. Hence a fixpoint variable may occur inside a modality's label, which §4 uses.

4 Derived forms

4.1 The spatial monoid

Proposition 4.1 ($\mathbf{0}$ is the unit of $|$). *Up to logical equivalence, $(\mathcal{F}, |, \mathbf{0})$ is a commutative monoid: $\phi | \mathbf{0} \equiv \phi$, $\phi | \psi \equiv \psi | \phi$, and $(\phi | \psi) | \chi \equiv \phi | (\psi | \chi)$. Moreover $|$ distributes over $\vee$ but not over $\wedge$, and $\mathbf{0} \equiv \neg(\neg \mathbf{0} | \text{true})$.*

Proof. For the unit, $P \models \phi | \mathbf{0}$ iff $P \equiv P_1 | P_2$ with $P_1 \models \phi$ and $P_2 \equiv \mathbf{0}$, iff $P \models \phi$. Commutativity and associativity descend from the abelian monoid structure of $\equiv$. Distribution over $\vee$ is immediate from the existential; failure over $\wedge$ is the usual one, witnessed by $\mathbf{m}(a, v) | \mathbf{m}(b, w)$ against $(\mathbf{m}(\text{@true}, \text{@true}) | \text{true}) \wedge (\mathbf{k}(\text{@true}) | \text{true})$. The last identity says that P is $\mathbf{0}$ iff it has no nonempty component. $\square$

4.2 Counting

Definition 4.2. For $k \geq 0$ and a formula θ , write

$$\begin{aligned} \#_\theta \geq k &\triangleq \underbrace{\theta | \dots | \theta}_k | \text{true}, & \#_\theta \leq k &\triangleq \neg(\#_\theta \geq k+1), \\ \#_\theta = k &\triangleq (\#_\theta \geq k) \wedge \neg(\#_\theta \geq k+1). \end{aligned}$$

Write NM for the disjunction of the structural productions of all shapes other than $\mathbf{m}$, each with every name argument @true and, for $\mathbf{q}$, payload true ; so $P \models \text{NM}$ iff P is a single non-message atom.

So $\#_{\text{NM}} \leq k$ is the atom-count measure of [7, Lem. 10.2], written as a formula. The counting of [5], which is primitive there, is here derived.

4.3 Nominals, and the characteristic formula

Definition 4.3 (Characteristic formula). For a closed term T define $\chi(T)$ by structural recursion:

$$\begin{aligned} \chi(\mathbf{0}) &= \mathbf{0} \\ \chi(T_1 | T_2) &= \chi(T_1) | \chi(T_2) \\ \chi(\alpha(\text{@}P_1, \dots, \text{@}P_n)) &= \alpha(\text{@}\chi(P_1), \dots, \text{@}\chi(P_n)) \quad (\alpha \neq \mathbf{q}) \\ \chi(\mathbf{q}(\text{@}P, S)) &= \mathbf{q}(\text{@}\chi(P), \chi(S)). \end{aligned}$$

Theorem 4.4 (Characteristic formulae are exact). *For all closed P and T : $P \models \chi(T)$ iff $P \equiv T$.*

Proof. Induction on $|T|$. The base case is Definition 3.6's clause for $\mathbf{0}$. For an atom, $P \models \alpha(\text{@}\chi(P_1), \dots)$ iff $P \equiv \alpha(a_1, \dots, a_n)$ with $a_i \models \text{@}\chi(P_i)$, iff by Definition 3.4 $a_i \equiv_{\mathcal{N}} \text{@}Q_i$ with $Q_i \models \chi(P_i)$, iff by induction $Q_i \equiv P_i$, iff $a_i \equiv_{\mathcal{N}} \text{@}P_i$ by Proposition 2.4. The case of $\mathbf{q}$ is the

same with the payload compared at the process sort. For $T_1 \mid T_2$, satisfaction of $\chi(T_1) \mid \chi(T_2)$ gives a split $P \equiv P_1 \mid P_2$ with $P_i \equiv T_i$ by induction, and conversely the split witnessing $P \equiv T_1 \mid T_2$ satisfies the clause. Each recursive call is on a proper subterm, so the induction is well founded. $\square$

Corollary 4.5 (Nominals). *The process nominal of [8, §6] and the singleton namespace of [5] are both derived: T as a formula is $\chi(T)$, and the singleton $\{a\}$ for $a = @P$ is the namespace $@\chi(P)$.*

Corollary 4.6 (Maximal intensionality). *$P \equiv Q$ iff P and Q satisfy the same formulae. The logic is therefore not adequate for any coarser equivalence, and in particular not for $\approx_{[\mathcal{C}_{\text{rho}}]}$.*

4.4 Temporal operators

Definition 4.7.

$$[\phi]\psi \triangleq \neg\langle\phi\rangle\neg\psi, \quad \Diamond\psi \triangleq \mu X(). \psi \vee \langle\mathbf{0}\rangle X, \quad \Box\psi \triangleq \neg\Diamond\neg\psi.$$

$\Diamond$ and $\Box$ range over internal reduction, since $\langle\mathbf{0}\rangle$ is the internal step by Remark 3.8; X occurs positively under $\langle\mathbf{0}\rangle$ by Proposition 3.11, so the fixpoint is well formed. The environment-facing analogue $\Diamond_\phi\psi \triangleq \mu X(). \psi \vee \langle\phi\rangle X$ is available and is a genuinely different operator, since it permits the context to be re-supplied at every step.

Definition 4.8 (Bounded termination). $\langle\mathbf{0}\rangle^0\phi \triangleq \phi$ and $\langle\mathbf{0}\rangle^{j+1}\phi \triangleq \langle\mathbf{0}\rangle\langle\mathbf{0}\rangle^j\phi$; and

$$\text{Halts}_k \triangleq \neg\langle\mathbf{0}\rangle^{k+1}\text{true},$$

which says no internal run from here has length exceeding k .

5 Model checking

5.1 The inhabitation relation

The relation to be decided is $T \models \phi$ for closed T and closed ϕ . We call it *inhabitation* when ϕ is read as a specification of a position in the lattice and *model checking* otherwise; they are the same relation.

Definition 5.1 (Hereditary material). $\text{Sub}^*(T)$ is the least set of $\equiv$ -classes containing T and closed under: if $P \in \text{Sub}^*(T)$ then every Q with $\text{comps}(Q) \subseteq \text{comps}(P)$ is in $\text{Sub}^*(T)$; and if $\alpha(a_1, \dots, a_n) \in \text{comps}(P)$ for some $P \in \text{Sub}^*(T)$, with $a_i = @P_i$, then each $P_i \in \text{Sub}^*(T)$, and for $\mathbf{q}(a, S)$ also $S \in \text{Sub}^*(T)$.

Lemma 5.2. *$\text{Sub}^*(T)$ is finite, of size at most $2^c \cdot n$ where $n = |T|$ and c is the maximum number of top-level components of any hereditary subterm of T .*

Proof. There are at most n hereditary subterms, and the sub-multiset closure of a term with c components contributes at most 2^c classes. Descent is by the subterm order, which is well founded on finite terms. $\square$

Theorem 5.3 (The modality-free fragment is decidable). *For closed T and closed modality-free ϕ , $T \models \phi$ is decidable.*

Proof. Every clause of Definition 3.6 other than the modality refers only to members of $\text{Sub}^*(T)$: the split clause to sub-multisets, the structural clauses to quoted subterms and, for $\mathbf{q}$, to the payload, and the name clause to the quoted process. By Lemma 5.2 the carrier is finite, so the operator $\Phi_{\mathcal{E}}$ of Definition 3.6 acts on the finite lattice $\mathcal{P}(\text{Sub}^*(T))^{\text{Sub}^*(T)^n}$ and its least fixpoint is reached by iteration in at most $|\text{Sub}^*(T)|^{n+1}$ steps. Namespace arguments range over $\{\text{@}P : P \in \text{Sub}^*(T)\}$, again finite. Evaluate bottom up. $\square$

Remark 5.4 (Cost). The bound of Lemma 5.2 is exponential in the number of parallel components, and the split clause is where that is spent: deciding $P \models \phi \mid \psi$ naively enumerates the $2^{|\text{comps}(P)|}$ splits. This is the familiar cost of a spatial composition connective and is not an artefact of this presentation. §6 shows that the formulae the lattice actually needs avoid it entirely.

5.2 The modal fragment

Proposition 5.5 (Decidability on the $\mathbf{e}$ -free sublattice, modulo witnesses). *Let $T \models \text{Occ}(\mathcal{A})$ for some rule-closed $\mathcal{A}$ with $\mathbf{e} \notin \mathcal{A}$. Then the reachable set of T is finite, and $T \models \phi$ is decidable for every closed ϕ whose modalities carry only labels ϕ_{ℓ} with $\llbracket \phi_{\ell} \rrbracket$ finite up to $\equiv$.*

Proof. By [7, Prop. 10.7] every reduction sequence from T has length at most $|\text{comps}(T)|_{\text{NM}}$, and by Definition 2.5 the branching at each step is bounded by the number of pairs and triples of components, so $\text{Reach}(T)$ is finite. Extend $\text{Sub}^*(-)$ over it; the argument of Theorem 5.3 then applies, with the modality clause evaluated by enumerating the finitely many $R \models \phi_{\ell}$ and, for each, the finitely many marked derivations, which are bounded by Remark 3.10. $\square$

The hypothesis on labels is the gap. A label such as $\mathbf{m}(\text{@}\chi(P), \text{@true})$ constrains the subject to one name and leaves the payload free, so $\llbracket \phi_{\ell} \rrbracket$ is infinite. The route to removing the hypothesis is a depth-bounded distinguishability lemma:

Conjecture 5.6 (Finite witnesses). *Define the quote depth $\text{qd}(\phi)$ of a μ -free formula as the maximum nesting of $\text{@}(-)$ in it. Then for μ -free ϕ_{ℓ} and ψ , satisfaction of $\langle \phi_{\ell} \rangle \psi$ at P is unchanged if R is restricted to range over a set of representatives of $\llbracket \phi_{\ell} \rrbracket$ of quote depth at most $\text{qd}(\phi_{\ell}) + \text{qd}(\psi) + |P|$, which is finite up to $\equiv$ once the names available are those of $\text{Sub}^*(P)$.*

The restriction to μ -free formulae is not cosmetic: unfolding a fixpoint raises quote depth without bound, so the bound must be re-derived per unfolding or the conjecture restated as a bisimulation-up-to-depth argument. This is listed as open item 1.

Proposition 5.7 (Undecidability above). *With $\mathbf{e}$ and $\mathbf{d}$ present the calculus is Turing complete [7, §7], and $\Diamond\chi(T')$ expresses reachability of T' . So $T \models \phi$ is undecidable for the full logic on the top vertex of the computation axis.*

Remark 5.8 (The decidability frontier is the destructor edge). Propositions 5.5 and 5.7 say that the logic is decidable exactly on the $\mathbf{e}$ -free half of the computation axis, which is the edge [7, Prop. 10.7] proves strict by strong normalisation. The occupancy formula for that edge is therefore also the certificate that model checking terminates. Should Conjecture 10.9 of [7] hold — that $\mathbf{d}$ -freedom gives strong normalisation even with $\mathbf{e}$ — the decidable region extends to that vertex too, so settling it settles where model checking stops working as well as where divergence starts.

6 Lattice inhabitation

6.1 Occupancy

Definition 6.1 (Hereditary atom set). $\text{at}^*(P)$ is the set of shapes α such that some atom of shape α occurs in P at some quote depth: the least set with $\alpha \in \text{at}^*(P)$ whenever $\alpha(\vec{a}) \in \text{comps}(P)$, and $\text{at}^*(Q) \subseteq \text{at}^*(P)$ whenever $@Q$ is an argument of such an atom or Q is the payload of a $\mathbf{q}$ in it.

Definition 6.2 (Occupancy formula). For $\mathcal{A} \subseteq \mathcal{A}_{\text{full}}$,

$$\text{Occ}(\mathcal{A}) \triangleq \mu X(). \mathbf{0} \vee (X \mid X) \vee \bigvee_{\alpha \in \mathcal{A} \setminus \{\mathbf{q}\}} \underbrace{\alpha(@X, \dots, @X)}_{\text{ar}(\alpha)} \vee [\mathbf{q}(@X, X) \text{ if } \mathbf{q} \in \mathcal{A}].$$

The formula is the atom list of Definition 2.1 with the disjuncts for shapes outside $\mathcal{A}$ deleted, plus the two spatial disjuncts. The $\mathbf{q}$ disjunct is the only one whose recursive occurrence appears at both sorts, which is Remark 3.2 again.

Proposition 6.3 (Occupancy is hereditary containment). $P \models \text{Occ}(\mathcal{A})$ iff $\text{at}^*(P) \subseteq \mathcal{A}$; equivalently, iff P is a term of the sub-calculus whose atom set is $\mathcal{A}$.

Proof. Right to left, the set of terms with $\text{at}^*(P) \subseteq \mathcal{A}$ is closed under the operator of Definition 6.2, hence contains its least fixpoint by Knaster–Tarski, and conversely each disjunct preserves the property. Left to right, observe that each recursive occurrence of X is under a structural production or under $\mid$, and that the argument of a structural production is a proper subterm; so the unfolding descends the subterm order on the finite term P and terminates, at which point every atom encountered has been matched against a disjunct of $\mathcal{A}$. $\square$

Corollary 6.4. For $\text{Occ}(\mathcal{A})$ the least and greatest fixpoints coincide.

Proof. The unfolding descends the subterm order, which is well founded on finite terms, so no infinite unfolding is consistent and ν adds nothing to μ . $\square$

Theorem 6.5 (Occupancy is linear). $T \models \text{Occ}(\mathcal{A})$ is decidable in time linear in $|T|$.

Proof. By Proposition 6.3 the test is $\text{at}^*(T) \subseteq \mathcal{A}$, which is one traversal of T descending into quotations. The split clause of Remark 5.4 costs nothing here because the disjunct $X \mid X$ makes $\text{Occ}(\mathcal{A})$ a homomorphic predicate: $P \mid Q \models \text{Occ}(\mathcal{A})$ iff both conjuncts do, so no enumeration of splits is required. $\square$

Theorem 6.5 is the requirement of §1 discharged. The occupancy formulae live in the modality-free fragment, so Theorem 5.3 already gives decidability; what Theorem 6.5 adds is that they land in the part of that fragment which avoids the exponential entirely.

6.2 Invariance, and a constraint on the lattice as drawn

Definition 6.6 (Rule-closed). $\mathcal{A}$ is *rule-closed* if

- (R1) $\mathbf{m} \in \mathcal{A}$;
- (R2) $\mathbf{fw} \in \mathcal{A}$ whenever $\mathcal{A}$ meets $\{\mathbf{b}_l, \mathbf{b}_r, \mathbf{s}\}$;
- (R3) $\text{cons}_m \in \mathcal{A} \Rightarrow \mathbf{m} \in \mathcal{A}$, $\text{cons}_d \in \mathcal{A} \Rightarrow \mathbf{d} \in \mathcal{A}$, and $\text{cons}_s \in \mathcal{A} \Rightarrow \mathbf{s} \in \mathcal{A}$.

Proposition 6.7 (Invariance of occupancy). *Let $\mathcal{A}$ be rule-closed. If $P \models \text{Occ}(\mathcal{A})$ and $P \longrightarrow P'$ then $P' \models \text{Occ}(\mathcal{A})$. Equivalently, $\text{Occ}(\mathcal{A}) \Rightarrow \Box \text{Occ}(\mathcal{A})$ is valid.*

Proof. By cases on Definition 2.5, using Proposition 6.3. The router rules produce $\mathbf{m}$ atoms and, for $\mathbf{b}_l$, $\mathbf{b}_r$, $\mathbf{s}$, an fw atom; (R1) and (R2) cover these. The rule for $\mathbf{e}$ produces the payload P of a message, whose atoms were already hereditarily present. The rule for $\mathbf{q}$ produces $\mathbf{m}(b, @P)$, with P the stored payload, again already present. A constructor rule for shape A produces $\mathbf{m}(c, @A(\vec{v}))$, whose atoms are those of $\vec{v}$ together with A itself; (R1) and (R3) cover the four cases, with cons_l needing nothing since parallel composition is not an atom. Structural congruence introduces no atoms. $\square$

Proposition 6.8 (The two axes are not independent for occupancy). *The lattice of [7, Fig. 2] is drawn as a product of a computation axis over $\{\mathbf{d}, \mathbf{e}\}$ with a constructor axis over the four constructors. At the eight points carrying cons_d without $\mathbf{d}$, the atom set is not rule-closed, and occupancy is not invariant.*

Proof. Take $\mathcal{A}$ with $\text{cons}_d \in \mathcal{A}$, $\mathbf{d} \notin \mathcal{A}$, and let

$$T = \text{cons}_d(a, b, c, e) \mid \mathbf{m}(a, u) \mid \mathbf{m}(b, v) \mid \mathbf{m}(c, w) \longrightarrow \mathbf{m}(e, @d(u, v, w)) = T'.$$

Then $\text{at}^*(T) \subseteq \mathcal{A}$ but $\mathbf{d} \in \text{at}^*(T')$, so $T \models \text{Occ}(\mathcal{A})$ and $T' \not\models \text{Occ}(\mathcal{A})$. The eight points are the two computation points omitting $\mathbf{d}$ crossed with the four constructor points including cons_d . $\square$

Remark 6.9 (Three ways out). Proposition 6.8 is a statement about the sub-calculus, not about the logic, and it is the logic's first return on the investment. A sub-calculus that can build the name $@d(u, v, w)$ but has no $\mathbf{d}$ atom is not closed under its own reduction, because the names of a sub-calculus are quotations of its own processes. Three repairs are available and the note does not choose between them: (i) keep $\mathbf{d}$ in the base for those points, so the constructor axis is a product only over the remaining three; (ii) read occupancy non-hereditarily, at the cost of Proposition 6.3 no longer saying “is a term of”; or (iii) accept the eight points as expressively meaningful but not reduction-closed, and state the product claim for the expressive ordering only. Option (i) matches the intent of [7, §10.4], where $\mathbf{d}$ is discussed as a source of copies rather than as a shape a name may carry.

6.3 Name closure: the constructor axis

Occupancy says which atoms may appear. The constructor axis wants something sharper: which *names* can ever appear, which is [7, Lem. 10.1]. That lemma is a namespace fixpoint, and its disjuncts are in bijection with the constructors present.

Definition 6.10 (Closure namespace). Let T be closed with $\mathcal{N}^*(T) = \{@Q_1, \dots, @Q_r\}$ and let $S \subseteq \{\text{cons}_l, \text{cons}_m, \text{cons}_d, \text{cons}_s\}$. Put

$$\begin{aligned} \text{Cl}_S(T) &\triangleq @(\mu X(). \kappa_S), \quad \text{where } \kappa_S = \bigvee_{i=1}^r \chi(Q_i) \\ &\quad \vee [X \mid X] && \text{if } \text{cons}_l \in S \\ &\quad \vee [\mathbf{m}(@X, @X)] && \text{if } \text{cons}_m \in S \\ &\quad \vee [\mathbf{d}(@X, @X, @X)] && \text{if } \text{cons}_d \in S \\ &\quad \vee [\mathbf{s}(@X, @X, @X)] && \text{if } \text{cons}_s \in S. \end{aligned}$$

Definition 6.11 (Confinement).

$$\text{Conf}(U) \triangleq \mu X(U). \mathbf{0} \vee (X(U) \mid X(U)) \vee \bigvee_{\alpha \in \mathcal{A}_{\text{full}} \setminus \{\mathbf{q}\}} \underbrace{\alpha(\theta_U, \dots, \theta_U)}_{\text{ar}(\alpha)} \vee \mathbf{q}(\theta_U, X(U)),$$

where $\theta_U \triangleq U \wedge @X(U)$ — a namespace, well formed by Proposition 3.5.

$\text{Conf}(U)$ says that every name occurring at any depth lies in U , and that the processes those names quote are themselves confined. This is the one place the parameterised fixpoint earns its keep: the parameter U must be carried unchanged through the recursion, and a nullary fixpoint cannot do it, because the namespace being tested is not definable from the term at hand.

Proposition 6.12 (Name closure, in the logic). *Let T be closed, S the set of constructors in $\mathcal{A}$, and $\mathcal{A}$ rule-closed. Then $T \models \text{Conf}(\text{Cl}_S(T))$, and $\text{Conf}(\text{Cl}_S(T)) \Rightarrow \Box \text{Conf}(\text{Cl}_S(T))$ is valid at T .*

Proof. The first claim is immediate: every name of T is some $@Q_i$, matched by the corresponding disjunct of Definition 6.10 via Theorem 4.4. The second is [7, Lem. 10.1] transcribed. Every rule other than a constructor rule has on its right only names occurring on its left; a constructor rule for shape A emits $\mathbf{m}(c, @A(\vec{v}))$ with $\vec{v}$ already present, and $@A(\vec{v})$ lies in $\text{Cl}_S(T)$ precisely because the disjunct for cons_A is present in Definition 6.10 exactly when $\text{cons}_A \in S$. $\square$

Corollary 6.13 (Root exclusion). *Write Root_A for the formula $\neg \mathbf{0} \mid \neg \mathbf{0}$ when A is parallel composition, and $A(@\text{true}, \dots, @\text{true})$ otherwise. If $\text{cons}_A \notin S$ and no name of $\mathcal{N}^*(T)$ satisfies $@\text{Root}_A$, then*

$$T \models \Box \neg (\mathbf{m}(@\text{Root}_A, @\text{true}) \mid \text{true}),$$

that is, no message on a subject of root A is ever emitted.

Proof. By Proposition 6.12 every name reachable from T lies in $\text{Cl}_S(T)$, whose defining disjunction has no clause with root A and whose base disjuncts are excluded by hypothesis. A message with such a subject would require one. $\square$

Corollary 6.13 is the converse half of [7, Prop. 10.5] in logical form. The forward half — that the behaviour *is* realised when cons_A is present — is a modal formula:

Definition 6.14 (Separation witness). Let A be a node shape of arity n , and let the names a , v and $k_2, \dots, k_n$ be fixed. Put

$$W_A \triangleq \langle \mathbf{m}(@\chi(P_a), @\chi(P_v)) \rangle \Diamond (\mathbf{m}(@\chi(A(v, k_2, \dots, k_n)), @\text{true}) \mid \text{true}),$$

where $a = @P_a$ and $v = @P_v$.

W_A is Spec_A of [7, Def. 10.4] written as a formula: on receiving v at a , emit a message on the constructed subject. By Corollary 6.13, W_A is unsatisfiable under $\text{Occ}(\mathcal{A}) \wedge \text{Conf}(\text{Cl}_S(T))$ when $\text{cons}_A \notin S$ and v is chosen outside $\mathcal{N}^*(T)$, and it is satisfiable when cons_A , $\mathbf{d}$, $\mathbf{b}_r$ and $\mathbf{m}$ are present. The choice of v remains a side condition of the separation argument rather than part of the formula, since it quantifies over the term being separated.

6.4 The formulae, point by point

Definition 6.15 (Base). $\mathcal{A}_0 \triangleq \{m, k, fw, b_l, b_r, s, q\}$: the medium, the store, and the five routers that are not d .

Definition 6.16 (Lattice points). For $c \subseteq \{d, e\}$ and $S \subseteq \{\text{cons}_l, \text{cons}_m, \text{cons}_d, \text{cons}_s\}$ with $S \neq \emptyset \Rightarrow \text{cons}_l \in S$, put $\mathcal{A}_{c,S} \triangleq \mathcal{A}_0 \cup c \cup S$, and

$$\text{At}_{c,S} \triangleq \text{Occ}(\mathcal{A}_{c,S}), \quad \text{In}_{c,S}(T) \triangleq \text{Occ}(\mathcal{A}_{c,S}) \wedge \text{Conf}(\text{Cl}_S(T)).$$

$\text{At}_{c,S}$ is the occupancy formula: it holds of exactly the terms of the sub-calculus. $\text{In}_{c,S}(T)$ adds the name-closure invariant, which is what the strictness arguments consume; it is term-relative because the closure namespace is generated from $\mathcal{N}^*(T)$.

axis	point	formula	by
computation	$e \notin c$	$\text{Occ}(\mathcal{A}_{c,S})$, e disjunct deleted	Prop. 6.3
	with $\#_{\text{NM}} \leq k$	$\Rightarrow \text{Halts}_k$	[7, Prop. 10.7]
	$d \notin c$	$\text{Occ}(\mathcal{A}_{c,S})$, d disjunct deleted	Prop. 6.3
constructors	$\text{cons}_A \notin S$	$\text{Occ}(\mathcal{A}_{c,S})$, cons_A disjunct deleted	Prop. 6.3
	name closure	$\text{Conf}(\text{Cl}_S(T))$, A disjunct deleted	Prop. 6.12
	separation	$\neg W_A$	Cor. 6.13
both	invariance	$\square$ of the above, $\mathcal{A}$ rule-closed	Prop. 6.7
	the eight bad points	$\text{cons}_d \in S$, $d \notin c$: not invariant	Prop. 6.8

Proposition 6.17 (The computation axis, in the logic). $\text{Occ}(\mathcal{A}_{c,S}) \wedge (\#_{\text{NM}} \leq k) \Rightarrow \text{Halts}_k$ is valid whenever $e \notin c$.

Proof. [7, Lem. 10.2]: every rule but that for e strictly decreases the number of non-message atoms, and $\#_{\text{NM}}$ counts exactly those by Definition 4.2. Hence no internal run exceeds k steps, which is Definition 4.8. $\square$

Remark 6.18 (What the formulae do not decide). $\text{At}_{c,S}$ decides whether T is a term of the sub-calculus. It does not decide whether T is *equivalent* to one, which is the lattice’s own ordering. The gap between the two is exactly the content of the strictness results: Proposition 6.17 and Corollary 6.13 are the bridges, and they run one way, ruling out behaviours below a point rather than characterising the terms at it. A characterisation of the semantic position — “ T is $\approx$ -equivalent to some term satisfying $\text{At}_{c,S}$ ” — is not expressible in this logic and is very unlikely to be decidable; see open item 4.

7 What the logic does not do

The observational lattice. [7, Fig. 1] is a second lattice over the same subsets, on which q and m occupy different points and which collapses against encoded observers. Its edges are receptivity statements: $m(a, v)$ is consumable by every consumer at a , while $q(a, P)$ is consumable only by fw [7, Rem. 3.6]. Receptivity is the $\forall\exists$ modality $\langle K \rangle\phi$ of [8, §7], and it is not derivable from $\langle\phi\rangle\psi$ and negation, which give only the $\exists\exists$ and $\forall\forall$ readings — may and must quantify over the same player. The logic as specified therefore sees the q – m distinction *structurally*, through

the separate productions $\mathbf{q}(\nu, \psi)$ and $\mathbf{m}(\nu_1, \nu_2)$, and not *behaviourally*. For occupancy that is enough. For anything about the observational ordering it is not, and the repair is to add $(\langle \phi \rangle \psi)$ as a second primitive.

Adequacy. By Corollary 4.6 the logic separates terms up to $\equiv$, so it is not a characteristic logic for $\approx_{[\mathcal{C}_{\text{rho}}]}$, nor for any coarser relation. This is a design choice and not an oversight: an occupancy formula must see the shape of a term, and any logic that does will be at least as discriminating as its structural layer. Identifying the extensional sublogic — the fragment invariant under $\approx_{\mathcal{D}}$ — is the natural companion question and is open item 3. The same tension in its general form is the subject of [4].

Minimal labels. The labels of $\langle \phi \rangle \psi$ are enabling contexts, following [7, §6]; the minimal-label discipline via relative pushouts is [1]’s and is not used here. Nothing in this note needs it, and Remark 3.10 means the enabling contexts are already small.

Cost. The logic has no grading. The name-growth question of [7, §13] — that with constructors the names are no longer subterms of the program — shows up here as the observation that $\text{Cl}_S(T)$ is an infinite namespace with a finite presentation, which is exactly the situation a graded clause is built for. That is the bridge to [8]’s stage two and is not pursued.

8 Open items

1. **Finite witnesses for the modality.** Conjecture 5.6, and its extension past the μ -free fragment. Without it Proposition 5.5 carries a hypothesis on labels that excludes the labels one actually writes.
2. **The eight non-invariant points.** Proposition 6.8 and Remark 6.9. This is a question for [7] rather than for the logic, and the answer changes [7, Fig. 2].
3. **The extensional sublogic.** Which formulae are invariant under $\approx_{[\mathcal{C}_{\text{rho}}]}$? The modality-only fragment is the obvious candidate and is not obviously closed under the fixpoint.
4. **Semantic position.** Is there any useful approximation to “equivalent to a term of the sub-calculus” that the logic can express? Remark 6.18.
5. **Receptivity.** Adding $(\langle \phi \rangle \psi)$, and whether Theorem 5.3 survives it. The $\forall$ over contexts is the same infinity that Decision 3.3 removed with the guarantee, so the answer is probably that it survives only under Conjecture 5.6.
6. **Implementation.** The structural fragment is a bottom-up tree-automaton run and should fall out of the existing `f1r3space` crate [5] with the restriction connectives deleted and the atom productions added. Theorem 6.5 says the occupancy checker is a separate and much simpler artefact, and it is the one the lattice needs.
7. **Duplication.** [7, Conj. 10.9]. Remark 5.8 gives a second reason to want it settled.

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